A PRIME IDEAL CONTAINING A PRODUCT OF TWO IDEALS CONTAINS ONE OF THEM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Primality of an ideal in element-wise form. Definition of a prime ideal, used at the last line of the counted proof (line 2776). Turns the membership of the product xy in the prime p into membership of one of the two factors.
- Membership in a product of ideals. The inclusion IJ contained in p, used at proof line 2775. Places the product of the two chosen elements inside the prime ideal.

Lemma 0.1. *Let R be a ring, I and J two ideals and $\mathfrak{p}$ a prime ideal containing the product IJ. Then $\mathfrak{p}$ contains I or J.*

Proof. Assume the contrary and take $x \in I \setminus \mathfrak{p}$ and $y \in J \setminus \mathfrak{p}$. Their product is an element of $IJ \subset \mathfrak{p}$, which contradicts the assumption that $\mathfrak{p}$ was prime. □

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